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State-Space Representation Theory: Canonical Forms

Many state models, one transfer function: the canonical realizations.

Ogata, *Modern Control Engineering*, Ch. 9.

In this tutorial you will:

- build the controllable, observable, and diagonal canonical forms,
- confirm they share the same poles and step response, and
- relate them through a similarity transformation.

Step through with **Ctrl+Enter**, or render a report with **publish**.

For a SISO transfer function

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_n}{s^n + a_1 s^{n-1} + \dots + a_n}$$

there are several standard state-space realizations, related to each other by a similarity transformation $\bar{x} = Tx$ (equivalently $A \rightarrow T^{-1}AT$, $B \rightarrow T^{-1}B$, $C \rightarrow CT$), which leaves the transfer function – and hence the input-output behavior – unchanged.

```
G = tf([1 6 12],[1 6 11 6]);  
[num,den] = tfdata(G,'v');  
fprintf('G(s) numerator: '); disp(num)  
fprintf('G(s) denominator: '); disp(den)
```

```
G(s) numerator:      0      1      6      12  
G(s) denominator:    1      6     11      6
```

Controllable Canonical Form

Useful for pole-placement design (the structure makes Ackermann's formula transparent). For a_1, a_2, a_3 the denominator coefficients (monic, $a_0 = 1$):

$$A_{cc} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & 1 & 0 & 0; & 0 & 1 & 0 \end{bmatrix}, \quad B_{cc} = \begin{bmatrix} 1; 0; 0 \end{bmatrix}$$

$$C_{cc} = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}$$

where each b_i is a numerator coefficient of the same order, after removing any b_0 direct term.

```

a1 = den(2); a2 = den(3); a3 = den(4);
b1 = num(2)-num(1)*a1;
b2 = num(3)-num(1)*a2;
b3 = num(4)-num(1)*a3;
A_cc = [-a1 -a2 -a3; 1 0 0; 0 1 0];
B_cc = [1;0;0];
C_cc = [b1 b2 b3];
D_cc = num(1);
sys_cc = ss(A_cc,B_cc,C_cc,D_cc);

```

Observable Canonical Form

The transpose structure of controllable canonical form – useful for observer (Luenberger) design by duality:

$$A_{oc} = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix}, \quad B_{oc} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \quad C_{oc} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

```

A_oc = [-a1 1 0; -a2 0 1; -a3 0 0];
B_oc = [b1;b2;b3];
C_oc = [1 0 0];
D_oc = num(1);
sys_oc = ss(A_oc,B_oc,C_oc,D_oc);

```

Diagonal (Modal) Canonical Form

When the poles $p_1, \dots, p_n$ are distinct, partial-fraction expansion

$$\frac{Y(s)}{U(s)} = b_0 + \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \frac{c_3}{s - p_3}$$

gives a diagonal A whose entries are the poles, decoupling the states – each $\dot{x}_i = p_i x_i + u$ evolves independently.

```

[r_pf,p_pf,k_pf] = residue(num,den);
A_diag = diag(p_pf);
B_diag = ones(length(p_pf),1);
C_diag = r_pf';
D_diag = sum(k_pf);
sys_diag = ss(A_diag,B_diag,C_diag,D_diag);

```

Verifying Equivalence

All three realizations – and the original ‘tf-derived ‘ss(G)’ – share the same transfer function and poles, confirming they are similar representations of the same input-output system.

```

fprintf('Poles -- controllable: '); disp(pole(sys_cc))
fprintf('Poles -- observable:   '); disp(pole(sys_oc))
fprintf('Poles -- diagonal:     '); disp(pole(sys_diag))

figure
hold on
step(sys_cc)
step(sys_oc,'--')
step(sys_diag,':')
hold off
legend('Controllable canonical','Observable canonical','Diagonal canonical', ...
    'Interpreter','latex','FontSize',14)
title('Equivalent Realizations: Identical Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$','Interpreter','latex','FontSize',20)

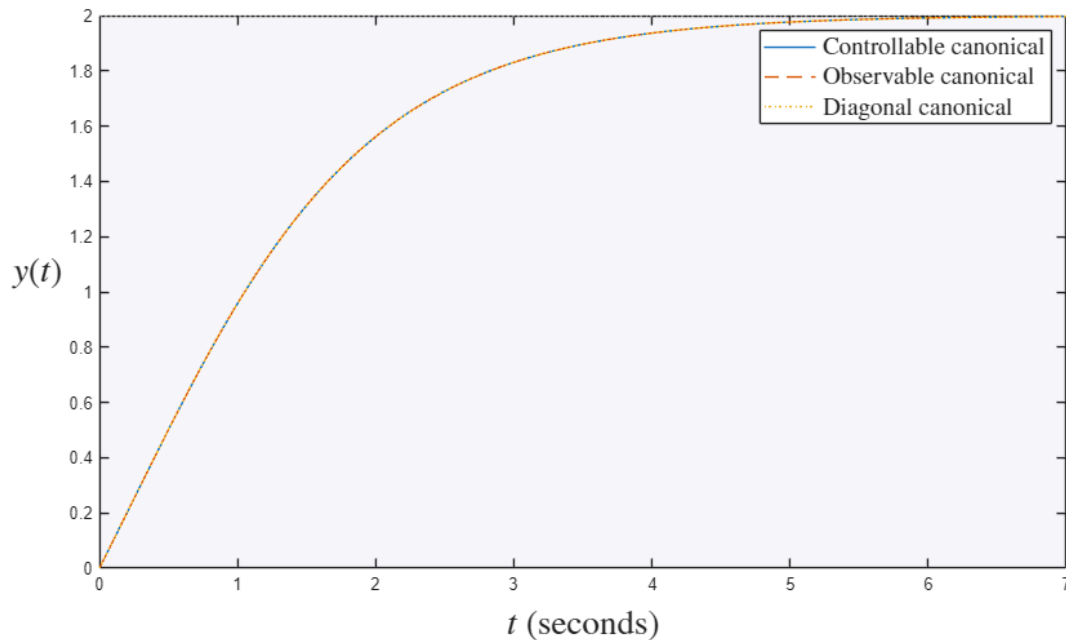
```

```

Poles -- controllable:  -3.0000  -2.0000  -1.0000
Poles -- observable:    -3.0000  -2.0000  -1.0000
Poles -- diagonal:      -3.0000  -2.0000  -1.0000

```

Equivalent Realizations: Identical Step Response



The Similarity Transformation Matrix

Between controllable and observable canonical form realizations of the **same** system, T can be found by matching the controllability matrices: $T = \mathbf{C}_{cc}\mathbf{C}_{oc}^{-1}$ is generally not applicable directly between distinct canonical types without going through a common basis; here we instead verify the diagonal form's transformation directly via the eigenvector matrix of A_{cc} .

```

[V_eig,~] = eig(A_cc);
T = V_eig;
A_check = T\A_cc*T;
fprintf('T^{-1}*A_cc*T (should match diag(poles), up to reordering):\n')
disp(real(A_check))

```

```

T^{-1}*A_cc*T (should match diag(poles), up to reordering):
-3.0000    0.0000    0.0000
 0.0000   -2.0000    0.0000
 0.0000   -0.0000   -1.0000

```

Try it yourself

- Give G a repeated pole and notice the diagonal (modal) form break down – **residue** needs distinct poles.
- Verify `tf(sys_cc)`, `tf(sys_oc)`, and `tf(sys_diag)` are all identical.